

A GENERAL MODEL FOR NEGATIVE MULTINOMIAL FREQUENCY COUNTS

H.S. Steyn: Universities and Technicons Advisory Council, Private Bag X122, Pretoria, 0001.

N.A.S. Crowther: Institute for Statistical Research, Human Sciences Research Council, Private Bag X41, Pretoria, 0001.

E.L. Raath: Department of Statistics, University of South Africa, P.O. Box 392, Pretoria, 0001.

and

T.M. Shaw: Institute for Statistical Research, Human Sciences Research Council, Private Bag X41, Pretoria, 0001.

Keywords: Categorical data; contingency tables; log linear model; asymptotic normality.

Summary: A general model for the analysis of frequency counts, where the sampling procedure is negative multinomial, is considered. In particular standard results for log-linear model analyses are derived. Estimation procedures for cell probabilities and log-linear parameters are developed for two way and three way classifications. A general analysis procedure, based on asymptotic theory, is also derived.

1. INTRODUCTION

The negative multinomial probability distribution, which is obtained when a random sampling procedure is terminated on obtaining a specified count in a predetermined cell (the target cell) is well-known in the field of discrete distributions associated with inverse sampling procedures. Amongst others, this distribution has proved to be of great value in ecological research when analysing data from capture-recapture experiments. In fact, it is stated by Taillie, Ord, Mosimann and Patil (1979, p.177) that a fixed target cell approach gives estimates with better sampling properties because of the guarantee that a predetermined number of tagged individuals will be recaptured.

However, in categorical data analysis, where frequency counts are displayed in a two-way or higher order contingency table, the existing log-linear procedures are based almost exclusively on an underlying multinomial model. One of the reasons why the negative multinomial has not been used up to now in log-linear analysis of categorical data is that the well-known forms of the chi-square expressions do not apply to the negative multinomial model. (Steyn 1955, 1959). A paper by Gokhale (1979, p.39) on the analysis of ecological frequency data using a multinomial model contains a number of interesting examples including data on recapture of whittings. In view of the comments by Taillie et al, referred to above, it can be expected that such experiments based on a negative multinomial model (that is, a fixed-target-cell model) may have distinct advantages over the multinomial model.

In human sciences research, it is often necessary to focus attention on a specific group of individuals, of which the probability of occurrence in an ordinary random sampling procedure may be small. In order to assure reasonable variances of the estimators of the log-linear parameters relating to this cell, a negative multinomial sampling procedure seems to be advisable.

Although the target cell will usually be the one believed to have the lowest probability of occurrence, this need not necessarily be the case. Each

particular experiment may present reasons for choosing a specific target cell.

The purpose of this paper is to derive the formulae to be used for analysing contingency tables obtained by a negative multinomial sampling procedure. The special case where each row of a two-way table has a negative multinomial distribution (i.e. all first column entries are fixed) has been dealt with by Rudolph (1963) and Bonett (1985).

2. THE NEGATIVE MULTINOMIAL MODEL

A simple model leading to the negative multinomial distribution is as follows: Consider the case of $r + 1$ mutually exclusive events E_i , $i = 0, 1, \dots, r$; and let p_i be the probability for an occurrence of E_i in a single trial, then, if the independent trials are terminated on obtaining the m -th success (including the last trial) of E_0 , the probability for obtaining $X_i = x_i$ occurrences of E_i , $i = 1, 2, \dots, r$; is given by,

$$(2.1) \quad P(X_1 = x_1, \dots, X_r = x_r) = \frac{(m + \sum_1^r x_i - 1)!}{(m - 1)! \prod_1^r x_i!} \left(\prod_1^r p_i^{x_i} \right) p_0^m.$$

This is the probability function of the negative multinomial distribution with parameters m and $\underline{p} = (p_1, p_2, \dots, p_r)'$ (Johnson and Kotz (1969), p.292)). The probability generating function (pgf) is

$$(2.2) \quad p_0^m (1 - \sum_1^r p_i t_i)^{-m}, \quad p_0 = 1 - \sum_1^r p_i.$$

Substituting $t_i = 1 + u_i$ in (2.2) to derive the factorial moment generating function ($fmgf$) and expanding this, the factorial moments of order $s_1 + s_2 + \dots + s_r$ follow as

$$(2.3) \quad \mu'_{(s_1, s_2, \dots, s_r)} = (m + \sum_1^r s_i - 1) (\sum_1^r s_i) \left(\prod_1^r p_i^{s_i} \right) / p_0^{\sum_1^r s_i}$$

where

$$z^{(a)} = z(z - 1) \cdots (z - a + 1)$$

so that

$$(2.4) \quad E(X_i) = mp_i/p_o, \quad Var(X_i) = \sigma_i^2 = mp_i(p_i + p_o)/p_o^2 \\ Cov(X_i, X_j) = \sigma_{ij} = mp_i p_j / p_o^2, i \neq j.$$

Substituting $t_j = 1, j = s+1, \dots, r$ the *pgf* (2.2) reduces to

$$\left(\frac{p_o}{p_o + \sum_{i=1}^s p_i} \right)^m \left(1 - \sum_{i=1}^s \frac{p_i}{p_o + \sum_{i=1}^s p_i} t_i \right)^{-m},$$

which shows that the marginal distribution of $X_1, X_2, \dots, X_s$ is negative multinomial with parameters m and $p_i/(p_o + \sum_{i=1}^s p_i)$, $i = 1, 2, \dots, s$.

Also, substituting

$$t_1 = t_2 = \dots = t_{s_1} = t'_1;$$

$$t_{s_1+1} = t_{s_1+2} = \dots = t_{s_2} = t'_2; \dots; t_{s_{h-1}+1} = \dots = t_r = t'_h;$$

it is clear from (2.2) that

$$E \left[\prod_{l=1}^h t_l^{Z_l} \right] = p_o^m (1 - \sum_{l=1}^h p'_l t'_l)^{-m}$$

where

$$Z_1 = \sum_{j=1}^{s_1} X_j, \quad Z_2 = \sum_{j=s_1+1}^{s_2} X_j, \quad \dots, \quad Z_h = \sum_{j=s_{h-1}+1}^r X_j$$

and where

$$p'_l = \sum_{j=s_{l-1}+1}^{s_l} p_j, \quad s_0 = 0, l = 1, 2, \dots, h.$$

The joint distribution of the linear sums $Z_l, l = 1, 2, \dots, h$ is then again a negative multinomial distribution with parameters m and p'_l , so that $E(Z_l) = mp'_l/p_o$.

The maximum likelihood estimators of the cell probabilities are obtained by maximizing (2.1) w.r.t. the p_i 's and are given by:

$$(2.5) \quad \hat{p}_i = \frac{x_i}{m + \sum_{j=1}^r x_j}, \quad \left(\frac{p_i}{p_o} \right) = \frac{x_i}{m}.$$

Furthermore, using two linear sums Z_l and $Z_k, l \neq k$, the maximum likelihood estimator,

$$(2.6) \quad \left(\frac{\widehat{p'_l}}{p_0 + p'_k} \right) = \frac{z_l}{m + z_k},$$

will often be used when estimating parameters from linear sums such as row and column totals in contingency tables. The estimator (2.6) follows from (2.5) by the well-known properties of maximum likelihood estimators, but can easily be derived independently by considering the marginal probability function of Z_l and Z_k as functions of

$$\alpha_l = p'_l/p_0 \text{ and } \alpha_k = p'_k/p_0 \text{ i.e.}$$

$$\frac{(m + z_l + z_k - 1)!}{(m - 1)!z_l!z_k!} \alpha_l^{z_l} \alpha_k^{z_k} (1 + \alpha_l + \alpha_k)^{-(m + z_l + z_k)},$$

so that on differentiation w.r.t. α_l and α_k follows that:

$$z_l(1 + \alpha_k) = \alpha_l(m + z_k), \quad z_k(1 + \alpha_l) = \alpha_k(m + z_l)$$

or

$$\left(\frac{\widehat{\alpha_l}}{1 + \alpha_k} \right) = \frac{z_l}{m + z_k} \text{ i.e. } \left(\frac{\widehat{p'_l}}{p_0 + p'_k} \right) = \frac{z_l}{m + z_k}.$$

In the analyses of categorical data, linear restraints on the frequency counts are of fundamental importance. The result in (2.6) shows that in such cases quotients of linear combinations of frequency counts will have to be used for estimating the unknown parameters p_i . In this respect the negative multinomial model differs fundamentally from the multinomial model. Asymptotic theory which is so familiar in the multinomial model is, therefore, introduced rather cautiously in section 5 after the more exact treatment in sections 3 and 4. Further interesting properties concerning marginal and conditional distributions can be found in Johnson and Kotz (1969) and Steyn (1955).

The asymptotic distribution relative to m is multivariate normal, $N(\underline{\mu}, V)$, where from (2.4):

$$\underline{\mu} = \frac{m}{p_0} \underline{p}, \quad V = (m/p_0^2)[\underline{p}\underline{p}' + p_0 D_p]$$

and $D_p = \text{diag}(p_1, p_2, \dots, p_r)$, so that it follows from the relevant quadratic form in the exponent that

$$(2.7) \quad Y^2 = \sum_1^r \frac{(x_i - mp_i/p_o)^2}{mp_i/p_o} - \frac{(m + \sum_1^r x_i - m/p_o)^2}{m/p_o}$$

is distributed as chi-square with r degrees of freedom. The expression (2.7) was used by Steyn (1959) for analysing two-way negative multinomial contingency tables under the hypothesis of independence. A more general approach including log-linear analyses, is however needed.

Using (2.7) as starting point it can easily be shown that the estimators defined in (2.6) are also modified χ^2 -minimum estimators (Cramér (1966) p.426)

3. NEGATIVE MULTINOMIAL TWO-WAY CONTINGENCY TABLES

Consider the following $I \times J$ contingency table constructed by a sampling process where the experiment was terminated on obtaining the m -th count in the target cell, which without loss of generality, may be taken as cell (1,1). The total frequency is therefore a random variable.

TABLE 3.1
OBSERVED FREQUENCY AND TOTALS

$f_{11} = m$	f_{12}	$\cdots$	f_{1j}	$\cdots$	f_{1J}	$f_{1+} = m + \sum_2^J f_{1j}$
f_{21}	f_{22}	$\cdots$	f_{2j}	$\cdots$	f_{2J}	$f_{2+} = \sum_1^J f_{2j}$
$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
f_{i1}	f_{i2}	$\cdots$	f_{ij}	$\cdots$	f_{iJ}	$f_{i+} = \sum_{j=1}^J f_{ij}$
$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
f_{I1}	f_{I2}	$\cdots$	f_{Ij}	$\cdots$	f_{IJ}	$f_{I+} = \sum_1^J f_{Ij}$
$f_{+1} =$	$f_{+2} =$	$\cdots$	$f_{+j} =$	$\cdots$	$f_{+J} =$	$f_{++} =$
$m + \sum_2^I f_{i1}$	$\sum_1^I f_{i2}$	$\cdots$	$\sum_{i=1}^I f_{ij}$	$\cdots$	$\sum_1^I f_{iJ}$	$m + \sum_1^I \sum_{j>2}^J f_{ij}$

The probability function associated with the above table is given by (2.1) as:

$$\frac{(m + \sum_{i+j>2}^I \sum^J f_{ij} - 1)!}{(m-1)! \prod_{i+j>2}^I \prod^J (f_{ij}!)} \left(\prod_{i+j>2}^I \prod^J p_{ij}^{f_{ij}} \right) p_{11}^m$$

where p_{ij} is the probability for a count in cell (i, j) in a single trial. Using the notation $p_{+j} = \sum_{i=1}^I p_{ij}$ and $p_{i+} = \sum_{j=1}^J p_{ij}$ it follows from (2.5) and (2.6) that:

$$\left(\frac{p_{ij}}{p_{11}} \right) = \frac{f_{ij}}{m}, \quad i+j > 2$$

$$(3.1) \quad \left(\frac{p_{1+}}{p_{11}} \right) = \frac{m + \sum_{j=2}^J f_{1j}}{m},$$

$$(3.2) \quad \left(\frac{p_{i+}}{p_{11}} \right) = \frac{f_{1+}}{m}, \quad \left(\frac{p_{+j}}{p_{11}} \right) = \frac{f_{+j}}{p_{11}}, \quad \left(\frac{1}{p_{11}} \right) = \frac{f_{++}}{m}$$

$$(3.3) \quad \left(\frac{p_{i+}}{p_{1+}} \right) = \frac{f_{i+}}{f_{1+}}, \quad \left(\frac{p_{+j}}{p_{+1}} \right) = \frac{f_{+j}}{f_{+1}}.$$

Also,

$$(3.4) \quad \text{var}(f_{1+}) = \text{var}\left(\sum_{j=2}^J f_{1j}\right) = mp_{1+}(p_{1+} - p_{11})/p_{11}^2,$$

$$\text{var}(f_{+1}) = mp_{+1}(p_{+1} - p_{11})/p_{11}^2,$$

and for $i > 1, j > 1$,

$$(3.5) \quad \text{var}(f_{i+}) = mp_{i+}(p_{i+} + p_{11})/p_{11}^2; \quad \text{var}(f_{+j})$$

$$= mp_{+j}(p_{+j} + p_{11})/p_{11}^2;$$

$$\text{cov}(f_{i+}, f_{r+}) = mp_{i+}p_{r+}/p_{11}^2, \quad i \neq r.$$

Writing $F_{ij} = E(f_{ij})$ the expected frequencies are given in Table 3.2.

TABLE 3.2
EXPECTED FREQUENCIES AND TOTALS

m	F_{12}	$\dots$	F_{1j}	$\dots$	F_{1J}	$F_{1+} = mp_{1+}/p_{11}$
F_{21}	F_{22}	$\dots$	F_{2j}	$\dots$	F_{2J}	$F_{2+} = mp_{2+}/p_{11}$
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$
F_{i1}	F_{i2}	$\dots$	F_{ij}	$\dots$	F_{iJ}	$F_{i+} = mp_{i+}/p_{11}$
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$
F_{I1}	F_{I2}	$\dots$	F_{Ij}	$\dots$	F_{IJ}	$F_{I+} = mp_{I+}/p_{11}$
$F_{+1} =$	$F_{+2} =$	$\dots$	$F_{+j} =$	$\dots$	$F_{+J} =$	$F_{++} = m+$
mp_{+1}/p_{11}	mp_{+2}/p_{11}		mp_{+j}/p_{11}		mp_{+J}/p_{11}	$\sum^I \sum_{i+j>2}^J F_{ij}$
						$= m/p_{11}$

4. A LOG-LINEAR MODEL

4.1 Definition of the model

Under the hypothesis of independence of rows and columns in Table 3.1, that is, $p_{ij} = p_{i+}p_{+j}$ for all (i, j) including $(1, 1)$, it follows that

$$(4.1) \quad F_{ij} = mp_{i+}p_{+j}/(p_{1+}p_{+1}) = F_{i+}F_{+j}/F_{++},$$

so that

$$(4.2) \quad \ln F_{ij} = \ln F_{i+} + \ln F_{+j} - \ln F_{++}.$$

For interaction between rows and columns

$$(4.3) \quad F_{ij} = [F_{i+}F_{+j}/F_{++}]p_{ij}/(p_{i+}p_{+j}), \quad i+j > 2$$

so that

$$(4.4) \quad \ln F_{ij} = \ln F_{i+} + \ln F_{+j} - \ln F_{++} + \ln [p_{ij}/(p_{i+}p_{+j})].$$

If $z_{ij} = \ln F_{ij}$, it follows immediately, by taking means and using the same notation as in Tables 3.1 and 3.2, that

$$(4.5) \quad \begin{aligned} z_{ij} &= \bar{z}_{++} + (\bar{z}_{i+} - \bar{z}_{++}) + (\bar{z}_{+j} - \bar{z}_{++}) \\ &\quad + (z_{ij} - \bar{z}_{i+} - \bar{z}_{+j} + \bar{z}_{++}) \\ &= \mu + \lambda_i^A + \lambda_j^B + \lambda_{ij}^{AB} \end{aligned}$$

where

$$(4.6) \quad \begin{aligned} \mu &= \frac{1}{IJ} \sum_1^I \sum_1^J \ln F_{ij} \text{ with } F_{11} = m, \\ \lambda_i^A &= \frac{1}{J} \sum_{j'} \ln F_{ij'} - \frac{1}{IJ} \sum_{i'} \sum_{j'} \ln F_{i'j'}, \\ \lambda_j^B &= \frac{1}{I} \sum_{i'} \ln F_{i'j} - \frac{1}{IJ} \sum_{i'} \sum_{j'} \ln F_{i'j'}, \\ \lambda_{ij}^{AB} &= \ln F_{ij} - \frac{1}{I} \sum_{i'} \ln F_{i'j} - \frac{1}{J} \sum_{j'} \ln F_{ij'} + \frac{1}{IJ} \sum_{i'} \sum_{j'} \ln F_{i'j'}. \end{aligned}$$

4.2 Estimation of the probabilities

For a log linear analysis, the expected values F_{ij} that appear in (4.1) - (4.6), have to be estimated from the observed frequencies in Table 3.1. For independence of rows and columns,

$$\hat{F}_{ij} = m \left(\frac{\widehat{p_{i+}}}{\widehat{p_{1+}}} \right) \left(\frac{\widehat{p_{+j}}}{\widehat{p_{+1}}} \right),$$

where from (3.3), the maximum likelihood estimators are:

$$\left(\frac{\widehat{p_{i+}}}{\widehat{p_{1+}}} \right) = \frac{f_{i+}}{f_{1+}}, \text{ and } \left(\frac{\widehat{p_{+j}}}{\widehat{p_{+1}}} \right) = \frac{f_{+j}}{f_{+1}}.$$

As an example, consider the observed table:

$m = 5$	6	11
8	11	19
13	17	30

which, for estimated expected values $\hat{F}_{ij} = m \left(\frac{f_{i+}}{f_{1+}} \right) \left(\frac{f_{+j}}{f_{+1}} \right)$ gives the table

$m = 5$	6, 54	11, 54
8, 64	11, 29	19, 93
13, 64	17, 83	31, 47

From section 2 it is known that the linear sums f_{1+} , f_{i+} , f_{+1} and f_{+j} are asymptotically normally distributed. The distribution of the quotient of normal variables has been dealt with extensively in literature (see for example Marsaglia (1965); Yatchew (1986)).

From these references, it is clear that the asymptotic form of quotients of normal variables presents difficult problems, especially since the size of a single prescribed cell frequency in this consideration plays a fundamental role. A simulation process has therefore been used in Section 7 to indicate the normality of the different estimators.

The expressions for the λ -values in (4.6) are however linear functions of the $\ln F_{ij}$ or, in practical applications, of the $\ln \hat{F}_{ij}$ so that the problem of the quotients of normal variables then disappear.

4.3 Determining the λ -values

From (4.2) and (4.3) it follows that for independence of rows and columns

$$z_{ij} = \ln F_{ij} = \mu + \lambda_i^A + \lambda_j^B$$

where μ , λ_i^A and λ_j^B are given in (4.6) and $F_{ij} = mp_{i+}p_{+j}/(p_{1+}p_{+1})$, $i + j > 2$.

By estimating λ_i^A from

$$\hat{\lambda}_i^A = \frac{1}{J} \sum_{j=1}^J \ln \hat{F}_{ij} - \frac{1}{IJ} \sum_{i'=1}^I \sum_{j=1}^J \ln \hat{F}_{i'j}, \quad F_{11} = m$$

where $\hat{F}_{ij} = mf_{i+}f_{+j}/(f_{1+}f_{+1})$,

$$\hat{F}_{i+} = m \frac{f_{i+}}{f_{1+}} \frac{f_{++}}{f_{+1}}, \quad \hat{F}_{+j} = m \frac{f_{+j}}{f_{+1}} \frac{f_{++}}{f_{+1}}, \quad \hat{F}_{++} = m \frac{f_{++}}{f_{1+}} \frac{f_{++}}{f_{+1}},$$

it follows after algebraic simplification that

$$(4.7) \quad \hat{\lambda}_i^A = \ln f_{i+} - \frac{1}{I} \sum_{i'=1}^I \ln f_{i'+}$$

and similarly

$$\hat{\lambda}_j^B = \ln f_{+j} - \frac{1}{J} \sum_{j'=1}^J \ln f_{+j'}.$$

4.4 The variances of the estimated λ -values

In the case of independence between rows and columns, the variance of $\hat{\lambda}_i^A$ follows from (4.7) as

$$(4.8) \quad \text{var} \left(\hat{\lambda}_i^A \right) = \text{var} \left(\ln f_{i+} - \frac{1}{I} \sum_{i'=1}^I \ln f_{i'+} \right).$$

The following variances and covariances, obtained from (3.4) and (3.5), are necessary for computing this variance:

$$(4.9) \quad \begin{aligned} \text{var}(\ln f_{i+}) &\cong \frac{1}{[E f_{i+}]^2} \text{var}(f_{i+}), \quad i > 1 \\ &= \frac{p_{11}^2}{m^2 p_{i+}^2} \frac{m p_{i+} (p_{i+} + p_{11})}{p_{11}^2} \\ &= \frac{1}{F_{++}} \left(\frac{1}{p_{i+}} + \frac{1}{p_{11}} \right); \end{aligned}$$

$$(4.10) \quad \begin{aligned} \text{var}(\ln f_{1+}) &\cong \frac{p_{11}^2}{m^2 p_{1+}^2} \frac{m(p_{1+} - p_{11})p_{1+}}{p_{11}^2} \\ &= \frac{1}{F_{++}} \left(\frac{1}{p_{11}} - \frac{1}{p_{1+}} \right); \end{aligned}$$

$$(4.11) \quad \begin{aligned} \text{cov}(\ln f_{i+}, \ln f_{r+}) &\cong \frac{p_{11}^2}{m^2 p_{i+} p_{r+}} \frac{m p_{i+} p_{r+}}{p_{11}^2}, \\ &= \frac{1}{F_{++}} \frac{1}{p_{11}} \left(= \frac{1}{m} \right); \\ &\quad \text{with } i > 1, r > 1, i \neq r \end{aligned}$$

$$\begin{aligned}
 (4.12) \quad \text{cov}(\ln f_{i+}, \ln f_{1+}) &\cong \frac{p_{11}^2}{m^2 p_{i+} p_{1+}} \text{cov}(f_{i+}, \sum_2^J f_{1j}) \\
 &= \frac{p_{11}^2}{m^2 p_{i+} p_{1+}} \cdot \frac{m p_{i+} (\sum_2^J p_{1j})}{p_{11}^2} \\
 &= \frac{p_{11}}{m} \cdot \frac{(p_{1+} - p_{11})}{p_{1+} p_{11}} \\
 &= \frac{1}{F_{++}} \left(\frac{1}{p_{11}} - \frac{1}{p_{1+}} \right).
 \end{aligned}$$

It is interesting to note that for the independence model

$$\begin{aligned}
 (4.13) \quad \text{cov} \left(\ln \frac{f_{i+}}{f_{1+}}, \ln \frac{f_{+j}}{f_{+1}} \right), \quad i > 1, j > 1, \\
 &= \text{cov}(\ln f_{i+}, \ln f_{+j}) - \text{cov}(\ln f_{i+}, \ln f_{+1}) - \text{cov}(\ln f_{+j}, \ln f_{1+}) \\
 &\quad + \text{cov}(\ln f_{1+}, \ln f_{+1}) \\
 &= \frac{1}{m} \left[(1 + p_{1+} p_{+1}) - (1 - \sum_{j'=2}^J p_{1+} p_{+j'}) \right. \\
 &\quad \left. - (1 - \sum_{i'=2}^I p_{+1} p_{i'+}) - \sum_{i'=2}^I \sum_{j'=2}^J p_{i'+} p_{+j'} \right] \\
 &= \frac{1}{m} \left[1 - \sum_{i'=1}^I \sum_{j'=1}^J p_{i'+} p_{+j'} \right] \\
 &= 0
 \end{aligned}$$

where it has to be remembered that in the above algebraic simplification, f_{ij} appears in both f_{i+} and f_{+j} so that $\text{var}(f_{ij})$ must be taken into consideration.

Now

$$\begin{aligned}
 (4.14) \quad \text{var}(\hat{\lambda}_i^A) &= \text{var} \left[\left(1 - \frac{1}{I}\right) \ln f_{i+} - \frac{1}{I} \sum_{i' \neq i}^I \ln f_{i'+} \right] \\
 &= \left(\frac{I-1}{I} \right)^2 \text{var}(\ln f_{i+}) + \frac{1}{I^2} \sum_{i' \neq i}^I \text{var}(\ln f_{i'+}) \\
 &\quad - \frac{I-1}{I} \frac{2}{I} \sum_{i' \neq i}^I \text{cov}(\ln f_{i+}, \ln f_{i'+}) \\
 &\quad + \frac{2}{I^2} \sum_{i' < i''} \text{cov}(\ln f_{i'+}, \ln f_{i''+}) \quad (i' \text{ and } i'' \neq i).
 \end{aligned}$$

(i) For $i \neq 1$:

By using (4.9)-(4.12) it now follows from (4.14) that

$$\begin{aligned}
 (4.15) \quad \text{var}(\hat{\lambda}_i^A) &\cong \frac{1}{F_{++}} \left[\frac{I-2}{I} \frac{1}{p_{i+}} + \frac{1}{I^2} \left\{ \frac{1}{p_{i+}} - \frac{1}{p_{1+}} \right. \right. \\
 &\quad + \sum_{i'=2, i' \neq i}^I \frac{1}{p_{i'+}} + 2(I-1) \frac{1}{p_{1+}} \\
 &\quad \left. \left. - 2(I-2) \frac{1}{p_{1+}} \right\} \right] \\
 &= \frac{1}{F_{++}} \left[\frac{I-2}{I} \frac{1}{p_{i+}} + \frac{1}{I^2} \sum_{i'=1}^I \frac{1}{p_{i'+}} \right].
 \end{aligned}$$

(ii) For $i = 1$:

From (4.9)-(4.12) it follows from (4.14) that

$$\begin{aligned}
 (4.16) \quad \text{var}(\hat{\lambda}_1^A) &\cong \frac{1}{F_{++}} \left[\left(\frac{I-1}{I} \right)^2 \left(\frac{1}{p_{11}} - \frac{1}{p_{1+}} \right) + \frac{1}{I^2} \sum_{i'=2}^I \left(\frac{1}{p_{i'+}} + \frac{1}{p_{11}} \right) \right. \\
 &\quad \left. - 2 \frac{I-1}{I^2} \sum_{i'=2}^I \left(\frac{1}{p_{11}} - \frac{1}{p_{1+}} \right) + \frac{2}{I^2} \sum_{i' < i'', i' \neq 1} \frac{1}{p_{11}} \right] \\
 &= \frac{1}{F_{++}} \frac{1}{I^2} \left[\sum_{i'=2}^I \frac{1}{p_{i'+}} - \frac{1}{p_{1+}} \{ (I-1)^2 - 2(I-1)^2 \} \right] \\
 &= \frac{1}{F_{++}} \left[\frac{1}{I^2} \sum_{i'=1}^I \frac{1}{p_{i'+}} + \frac{I-2}{I} \frac{1}{p_{1+}} \right]
 \end{aligned}$$

so that the variance is the same for all $i = 1, 2, \dots, I$.

Similarly,

$$(4.17) \quad \text{var}(\hat{\lambda}_j^B) = \frac{1}{F_{++}} \left[\frac{J-2}{J} \frac{1}{p_{+j}} + \frac{1}{J^2} \sum_{j'=1}^J \frac{1}{p_{+j'}} \right].$$

For the saturated model, the variances of $\hat{\lambda}_i^A$, $\hat{\lambda}_j^B$ and $\hat{\lambda}_{ij}^{AB}$, defined in (4.6), but with F_{ij} replaced by $\hat{F}_{ij} = f_{ij}$, are obtained by using the following results:

$$\begin{aligned}
 (4.18) \quad \text{var}(\ln f_{ij}) &\cong \frac{1}{[E(f_{ij})]^2} \text{var}(f_{ij}), \quad (i, j) \neq (1, 1) \\
 &= \frac{p_{11}^2}{m^2 p_{ij}^2} \frac{m p_{ij} (p_{ij} + p_{11})}{p_{11}^2} \\
 &= \frac{1}{F_{++}} \left(\frac{1}{p_{ij}} + \frac{1}{p_{11}} \right)
 \end{aligned}$$

and

$$\begin{aligned}
 (4.19) \quad \text{cov}(\ln f_{ij}, \ln f_{rs}) &\cong \frac{1}{E(f_{ij})E(f_{rs})} \text{cov}(f_{ij}, f_{rs}), \quad (i, j) \neq (r, s) \\
 &= \frac{1}{F_{++}} \frac{1}{p_{11}}.
 \end{aligned}$$

$\hat{\lambda}_i^A$ for the saturated model follows as

$$(4.20) \quad \hat{\lambda}_i^A = \frac{1}{J} \sum_{j'=1}^J \ln f_{ij'} - \frac{1}{IJ} \sum_{i'=1}^I \sum_{j'=1}^J \ln f_{i'j'}.$$

(i) For $i \neq 1$, (4.20) can be written as

$$(4.21) \quad \begin{aligned} \hat{\lambda}_i^A &= \frac{1}{J} (\ln f_{i1} + \ln f_{i2} + \cdots + \ln f_{iJ}) - \frac{1}{IJ} (\ln m + \sum_{i'+j'>2} \sum_{i'=1}^I \sum_{j'=1}^J \ln f_{i'j'}) \\ &= \left(\frac{1}{J} - \frac{1}{IJ} \right) \sum_{j'=1}^J \ln f_{ij'} - \frac{1}{IJ} (\ln m + \sum_{i' \neq i, i'+j'>2} \sum_{i'=1}^I \sum_{j'=1}^J \ln f_{i'j'}) \end{aligned}$$

so that

$$(4.22) \quad \begin{aligned} \text{var}(\hat{\lambda}_i^A) &\cong \frac{1}{F_{++}} \left[\left(\frac{1}{J} - \frac{1}{IJ} \right)^2 \sum_{j'=1}^J \frac{1}{p_{ij'}} + \left(\frac{1}{IJ} \right)^2 \right. \\ &\quad \left. \sum_{i' \neq i, i'+j'>2} \sum_{i'=1}^I \sum_{j'=1}^J \frac{1}{p_{i'j'}} + \left(\frac{1}{IJ} \right)^2 \frac{1}{p_{11}} \right] \\ &= \frac{1}{F_{++}} \left[\left(\frac{1}{IJ} \right)^2 \sum_{i'=1}^I \sum_{j'=1}^J \frac{1}{p_{i'j'}} + \frac{I-2}{IJ^2} \sum_{j'=1}^J \frac{1}{p_{ij'}} \right]. \end{aligned}$$

(ii) For $i = 1$ (4.20) can be written as

$$\begin{aligned} \hat{\lambda}_1^A &= \frac{1}{J} (\ln m + \ln f_{12} + \cdots + \ln f_{1J}) - \frac{1}{IJ} \left(\sum_{i'+j'>2} \sum_{i'=1}^I \sum_{j'=1}^J \ln f_{i'j'} \right) \\ &= \left(\frac{1}{J} - \frac{1}{IJ} \right) \sum_{j'=2}^J \ln f_{1j'} - \frac{1}{IJ} \sum_{i' \neq 1}^I \sum_{j'=1}^J \ln f_{i'j'} + \frac{1}{J} \ln m. \end{aligned}$$

The procedure to find $\text{var}(\hat{\lambda}_1^A)$ is the same as that for case (i), except that the number of variables after the first and second summation sign now are $(J-1)$ and $(IJ-J)$ respectively, instead of J and $(IJ-J-1)$ as in case (i). Algebraic simplification again leads to (4.22) with $i = 1$. Similarly, $\text{var}(\hat{\lambda}_j^B)$ follows from (4.22) as

$$(4.23) \quad \text{var}(\hat{\lambda}_j^B) \cong \frac{1}{F_{++}} \left[\left(\frac{1}{IJ} \right)^2 \sum_{i'=1}^I \sum_{j'=1}^J \frac{1}{p_{i'j'}} + \frac{J-2}{I^2 J} \sum_{i'=1}^I \frac{1}{p_{i'j}} \right]$$

while it follows from (4.22) and (4.23) that

$$(4.24) \quad \text{var}(\hat{\lambda}_{ij}^{AB}) \cong \frac{1}{F_{++}} \left[\left(\frac{1}{IJ} \right)^2 \sum_{i'=1}^I \sum_{j'=1}^J \frac{1}{p_{i'j'}} \right. \\ \left. + \frac{I-2}{IJ^2} \sum_{j'=1}^J \frac{1}{p_{ij'}} + \frac{J-2}{I^2J} \sum_{i'=1}^I p_{i'j} \right. \\ \left. + \frac{(I-2)(J-2)}{IJ} \frac{1}{p_{ij}} \right].$$

The proof of (4.24) follows the same procedure as that for (4.22), but with careful handling of the number of combinations of covariances.

5. THE $I \times J \times K$ CONTINGENCY TABLE WITH A SPECIFIED COUNT IN CELL $(1, 1, 1)$

5.1 The log-linear model

Similarly to the two-way contingency table, it again follows that if f_{ijk} is the cell frequency in cell (i, j, k) , $i = 1, 2, \dots, I$; $j = 1, 2, \dots, J$; $k = 1, 2, \dots, K$, with $f_{111} = m$ and $E(f_{ijk}) = F_{ijk} = \frac{mp_{ijk}}{p_{111}}$, $z_{ijk} = \ln F_{ijk}$, then for the saturated model, by a logical extension of the previous notation in Section 4.1,

$$(5.1) \quad z_{ijk} = \bar{z}_{+++} + (\bar{z}_{i++} - \bar{z}_{+++}) + (\bar{z}_{+j+} - \bar{z}_{+++}) + (\bar{z}_{++k} - \bar{z}_{+++}) \\ + (\bar{z}_{ij+} - \bar{z}_{i++} - \bar{z}_{+j+} + \bar{z}_{+++}) \\ + (\bar{z}_{i+k} - \bar{z}_{i++} - \bar{z}_{++k} + \bar{z}_{+++}) \\ + (\bar{z}_{+jk} - \bar{z}_{+j+} - \bar{z}_{++k} - \bar{z}_{+++}) \\ + (z_{ijk} - \bar{z}_{ij+} - \bar{z}_{i+k} - \bar{z}_{+jk} + \bar{z}_{i++} + \bar{z}_{+j+} + \bar{z}_{++k} - \bar{z}_{+++}) \\ = \mu + \lambda_i^A + \lambda_j^B + \lambda_k^C + \lambda_{ij}^{AB} + \lambda_{ik}^{AC} + \lambda_{jk}^{BC} + \lambda_{ijk}^{ABC}.$$

In terms of the F_{ijk} 's, for example,

$$\lambda_i^A = \frac{1}{JK} \sum_{j'=1}^J \sum_{k'=1}^K \ln F_{ij'k'} - \frac{1}{IJK} \sum_{i'=1}^I \sum_{j'=1}^J \sum_{k'=1}^K \ln F_{i'j'k'}$$

so that an estimate for the λ_i^A is given by:

$$\hat{\lambda}_i^A = \frac{1}{JK} \sum_{j'} \sum_{k'} \ln f_{ij'k'} - \frac{1}{IJK} \sum_{i'} \sum_{j'} \sum_{k'} \ln f_{i'j'k'}.$$

As before

$$(5.2) \quad \text{var}(\hat{\lambda}_i^A) = \text{var} \left[\frac{1}{JK} \sum_{j'} \sum_{k'} \ln f_{ij'k'} - \frac{1}{IJK} \sum_{i'} \sum_{j'} \sum_{k'} \ln f_{i'j'k'} \right]$$

where

$$\begin{aligned} (5.3) \quad \text{var}(\ln f_{ijk}) &\cong \frac{1}{[E(f_{ijk})]^2} \text{var}(f_{ijk}), \quad (i, j, k) \neq (1, 1, 1) \\ &= \left[\frac{p_{111}}{mp_{ijk}} \right]^2 \frac{mp_{ijk}(p_{ijk} + p_{111})}{p_{111}^2} \\ &= \frac{1}{F_{+++}} \left(\frac{1}{p_{ijk}} + \frac{1}{p_{111}} \right) \end{aligned}$$

with

$$F_{+++} = E(m + \sum_{i+} \sum_{j+} \sum_{k>3} f_{ijk}) = \frac{m}{p_{111}}.$$

Furthermore

$$\begin{aligned} (5.4) \quad \text{cov}(\ln f_{ijk}, \ln f_{rst}) &\cong \frac{1}{E(f_{ijk})E(f_{rst})} \text{cov}(f_{ijk}, f_{rst}), \\ &\quad (i, j, k) \neq (r, s, t), \\ &= \frac{p_{111}^2}{mp_{ijk}mp_{rst}} \frac{mp_{ijk}p_{rst}}{p_{111}^2} \\ &= \frac{1}{F_{+++}} \frac{1}{p_{111}} \quad (= \frac{1}{m}). \end{aligned}$$

It is now clear that, by using the results (5.3) and (5.4) to determine (5.2) in terms of the p_{ijk} 's, this will be similar to the transition of (4.21) and (4.22) for the $I \times J$ table, both in the calculation of the contribution of the variances as well as the contribution of the covariances, but the number of linear sums appearing will increase from J to JK , so that the coefficient of $\frac{1}{F_{+++}} \frac{1}{p_{111}}$ will change to $\left(\frac{1}{IJK}\right)^2$. Consequently,

$$(5.5) \quad \text{var}(\hat{\lambda}_i^A) \cong \frac{1}{F_{+++}} \left[\left(\frac{1}{IJK}\right)^2 \sum_{i'=1}^I \sum_{j'=1}^J \sum_{k'=1}^K \frac{1}{p_{i'j'k'}} + \frac{I-2}{IJ^2K^2} \sum_{j'=1}^J \sum_{k'=1}^K \frac{1}{p_{ij'k'}} \right].$$

The result in (5.5) indicates that the variances for the λ coefficients as defined in (5.1) can be obtained by algebraic procedures, similar to the $I \times J$ case. The relevant variances are given in (5.6) and (5.7). The detailed derivations are given in a research report (1987) by the authors.

$$(5.6) \quad \text{var}(\hat{\lambda}_{ij}^{AB}) \cong \frac{1}{F_{+++}} \left[\left(\frac{1}{IJK}\right)^2 \sum_{i'=1}^I \sum_{j'=1}^J \sum_{k=1}^K \frac{1}{p_{i'j'k}} + \frac{I-2}{IJ^2K^2} \sum_{j'=1}^J \sum_{k=1}^K \frac{1}{p_{ij'k}} + \frac{J-2}{I^2JK^2} \sum_{i'=1}^I \sum_{k=1}^K \frac{1}{p_{i'jk}} + \frac{(I-2)(J-2)}{IJK^2} \sum_{k=1}^K \frac{1}{p_{ijk}} \right].$$

$$\begin{aligned}
 (5.7) \text{var}(\hat{\lambda}_{ijk}^{ABC}) \cong & \frac{1}{F_{+++}} \left[\left(\frac{1}{IJK} \right)^2 \sum_{i'=1}^I \sum_{j'=1}^J \sum_{k'=1}^K \frac{1}{p_{i'j'k'}} \right. \\
 & + \frac{I-2}{IJ^2K^2} \sum_{j'=1}^J \sum_{k'=1}^K \frac{1}{p_{ij'k'}} + \frac{J-2}{I^2JK^2} \sum_{i'=1}^I \sum_{k'=1}^K \frac{1}{p_{i'jk'}} \\
 & + \frac{K-2}{I^2J^2K} \sum_{i'=1}^I \sum_{j'=1}^J \frac{1}{p_{i'j'k}} \\
 & + \frac{(I-2)(J-2)}{IJK^2} \sum_{k'=1}^K \frac{1}{p_{ijk'}} + \frac{(I-2)(K-2)}{IJ^2K} \sum_{j'=1}^J \frac{1}{p_{ij'k}} \\
 & + \frac{(J-2)(K-2)}{I^2JK} \sum_{i'=1}^I \frac{1}{p_{i'jk}} \\
 & \left. + \frac{(I-2)(J-2)(K-2)}{IJK} \frac{1}{p_{ijk}} \right].
 \end{aligned}$$

The expressions (4.15), (4.22), (4.24), (5.5), (5.6) and (5.7) may be compared with the variances for the multinomial case as derived by Stoker and Glas (1984) where the total frequencies are fixed.

5.2 Estimating the expected frequencies

For the ordinary (positive) multinomial model there are a variety of null hypotheses that have to be tested. A list of the relevant hypotheses for the $I \times J \times K$ table appears in Stoker and Glas (1984) p.11. For example, the hypothesis

$$H_0 : \lambda_{ijk}^{ABC} = \lambda_{jk}^{BC} = \lambda_{ik}^{AC} = 0 \quad (\text{for all } i, j \text{ and } k),$$

means that for the testing λ_{ij}^{AB} , the classification can be collapsed over C . However, the same procedure holds for the $I \times J \times K$ table under a negative multinomial model since, if $H_0 : \lambda_{ijk}^{ABC} = \lambda_{jk}^{BC} = \lambda_{ik}^{AC} = 0$, for all $i + j + k > 1$, the expression for $\hat{\lambda}_{ij}^{AB}$ can, after collapsing over C , be rewritten as:

$$\hat{\lambda}_{ij}^{AB} = \ln f_{ij+} - \frac{1}{I} \sum_{i'} \ln f_{i'j+} - \frac{1}{J} \sum_{j'} \ln f_{ij'+} + \frac{1}{IJ} \sum_{i'} \sum_{j'} \ln f_{i'j'+}$$

from which it follows that

$$(5.8) \quad \text{var}(\hat{\lambda}_{ij}^{AB}) \cong \frac{1}{F_{+++}} \left\{ \left(\frac{1}{IJ} \right)^2 \sum_{i'} \sum_{j'} \frac{1}{p_{i'j'+}} + \frac{I-2}{IJ^2} \sum_{j'} \frac{1}{p_{ij'+}} \right. \\ \left. + \frac{J-2}{I^2J} \sum_{i'} \frac{1}{p_{i'j+}} + \frac{(I-2)(J-2)}{IJ} \frac{1}{p_{ij+}} \right\}.$$

The following hypotheses, expressed in terms of the cell probabilities, are usually considered (see Stumpf and Steyn (1986)):

- (5.9) (i) $p_{ijk} = 1/IJK$
- (ii) $p_{ijk} = p_{i++}/JK$
- (iii) $p_{ijk} = p_{ij+}/K$
- (iv) $p_{ijk} = p_{i++}p_{j+}/K$
- (v) $p_{ijk} = p_{i++}p_{j+}p_{++k}$
- (vi) $p_{ijk} = p_{ij+}p_{++k}$
- (vii) $p_{ijk} = p_{i+k}p_{+jk}/p_{++k}$.

The H_0 as defined above is the same as (vi). Estimation of the cell probabilities under the model of no second-order interactions, where no closed form of the estimates exist, are dealt with in Section 6 (see Section 6.3).

For the application of log linear analysis under the negative multinomial model, it is necessary to determine $\hat{F}_{ijk}$ and $\hat{F}_{+++}$, the estimated values of F_{ijk} and F_{+++} , for the seven hypotheses, so that the standardised values of the $\hat{\lambda}$'s, namely $\hat{\lambda}/\sqrt{\text{var}(\hat{\lambda})}$ can be determined for hypothesis testing under the assumption of normality.

Under the hypothesis (vi) above, for example, it follows for $i + j + k > 3$

$$F_{ijk} = m \frac{p_{ijk}}{p_{111}} = m \frac{p_{ij+} + p_{++k}}{p_{11+} + p_{++1}}, \quad F_{+++} = \frac{m}{p_{111}} = \frac{m}{p_{11+} + p_{++1}},$$

so that, from (2.6)

$$(5.10) \quad \begin{aligned} \hat{F}_{ijk} &= m \left(\frac{f_{ij+}}{f_{11+}} \right) \left(\frac{f_{++k}}{f_{++1}} \right) \\ \hat{F}_{+++} &= m \left(\frac{f_{+++}}{f_{11+}} \right) \left(\frac{f_{+++}}{f_{++1}} \right). \end{aligned}$$

Similarly, for the other hypotheses in (5.9), it follows that $\hat{F}_{ijk}$ and $\hat{F}_{+++}$ are given by

$$(i) \quad \hat{F}_{ijk} = m; \quad \hat{F}_{+++} = IJKm$$

$$(ii) \quad \hat{F}_{ijk} = m \frac{f_{i++}}{f_{1++}}; \quad \hat{F}_{+++} = m \frac{f_{+++}}{p_{1++}} JK$$

$$(iii) \quad \hat{F}_{ijk} = m \frac{f_{ij+}}{f_{11+}}; \quad \hat{F}_{+++} = m \frac{f_{+++}}{f_{11+}} K$$

$$(iv) \quad \hat{F}_{ijk} = m \frac{f_{i++} f_{+j+}}{f_{1++} f_{+1+}}; \quad \hat{F}_{+++} = m \left(\frac{f_{+++}}{f_{1++}} \right) \left(\frac{f_{+++}}{f_{+1+}} \right) K$$

$$(v) \quad \hat{F}_{ijk} = m \frac{f_{i++} f_{+j+} f_{++k}}{f_{1++} f_{+1+} f_{++1}}; \quad \hat{F}_{+++} = m \frac{f_{+++}}{f_{1++}} \frac{f_{+++}}{f_{+1+}} \frac{f_{+++}}{f_{++1}}$$

$$(vii) \quad \hat{F}_{ijk} = m \frac{f_{i+k} f_{+jk} f_{++1}}{f_{+1+} f_{+11} f_{++k}}; \quad \hat{F}_{+++} = m \frac{f_{+++} f_{+++}}{f_{+1+} f_{+11}}.$$

6. AN ESTIMATION PROCEDURE BASED ON ASYMPTOTIC THEORY.

6.1 Estimation Procedure

A general estimation procedure for the parameters in categorical data analysis, as described in Crowther and Shaw (1989), can be adapted in the case of negative multinomial sampling. As will be shown in Section 7, any frequency vector $\underline{f}' = (f_0, f_1, \dots, f_r) = (m, \underline{f}_1')$, observed according to a negative multinomial model, can be assumed to be approximately normal $N(\underline{F}, \Sigma)$, even for moderate values of m , where

$$(6.1) \quad \underline{F} = \frac{m}{p_0} \underline{p} = \begin{pmatrix} m \\ \underline{F}_1 \end{pmatrix}$$

and

$$(6.2) \quad \Sigma = \begin{pmatrix} 0 & 0' \\ 0 & D_{F_1} + \frac{1}{m} \underline{F}_1 \underline{F}_1' \end{pmatrix} = D_F + \frac{1}{m} \underline{F} \underline{F}' - (\underline{F}, 0) - \begin{pmatrix} \underline{F}' \\ 0 \end{pmatrix}$$

where $D_{F_1} = \text{diag}(\underline{F}_1)$ and $D_F = \text{diag}(\underline{F})$.

Any constraints on the elements of $\underline{F}$, including those specified by different hypotheses, can be expressed in the form

$$(6.3) \quad \underline{g}(\underline{F}) = \underline{c}$$

where $\underline{g}(\underline{f})$ is approximately normal. Let $G = \frac{\partial \underline{g}(\underline{F})}{\partial \underline{F}}$ be the matrix of partial derivatives of $\underline{g}(\underline{F})$ with respect to $\underline{F}$. The joint distribution of $\begin{pmatrix} \underline{f} \\ \underline{g}(\underline{f}) \end{pmatrix}$, under certain regularity conditions, is approximately normal with mean $\begin{pmatrix} \underline{F} \\ \underline{g}(\underline{F}) \end{pmatrix}$ and covariance matrix

$$\frac{\partial \begin{pmatrix} \underline{F} \\ \underline{g} \end{pmatrix}}{\partial \underline{F}} \Sigma \left(\frac{\partial \begin{pmatrix} \underline{F} \\ \underline{g} \end{pmatrix}}{\partial \underline{F}} \right)' = \begin{pmatrix} \Sigma & (G\Sigma)' \\ G\Sigma & G\Sigma G' \end{pmatrix}.$$

The conditional distribution of $\underline{f}$, given that $\underline{g}(\underline{f}) = \underline{c}$, is therefore approximately normal with expected value

$$(6.4) \quad \underline{F}_c = E(\underline{f} | \underline{g}(\underline{f}) = \underline{c}) = \underline{F} + (G\Sigma)'(G\Sigma G')^*(\underline{c} - \underline{g}(\underline{F}))$$

and covariance matrix

$$(6.5) \quad \Sigma_c = \Sigma - (G\Sigma)'(G\Sigma G')^*G\Sigma$$

where $(G\Sigma G')^*$ is any generalized inverse of $G\Sigma G'$. It follows too that

$$(6.6) \quad \text{rank}(\Sigma_c) = \text{rank}(\Sigma) - \text{rank}(G\Sigma).$$

Expression (6.4) specifies a general iterative estimation procedure for $\underline{F}$, with asymptotic covariance matrix of the estimator given by (6.5).

6.2 The log-linear model

For the log-linear modelling of contingency tables, the model can be written in general as

$$(6.7) \quad \ln \underline{F} = A\lambda$$

$$= (\underline{1}, A_1, A_2, \dots) \begin{pmatrix} \mu \\ \lambda^{A_1} \\ \lambda^{A_2} \\ \vdots \end{pmatrix}$$

where the design matrix A consists of the submatrices $\underline{1}, A_1, A_2, \dots$ corresponding to the mean, independent main effects, independent first order interactions, etc. Any standard hypothesis is of the form

$$(6.8) \quad \underline{g}(\underline{F}) = A'_H \ln \underline{F} = \underline{0}$$

where A_H consists of a subset of the A_i 's matrices. The matrix of partial derivatives is

$$(6.9) \quad G = \frac{\partial \underline{g}}{\partial \underline{F}} = A'_H D_F^{-1}.$$

Since $A'_H D_F^{-1} \underline{F} = A'_H \underline{1} = \underline{0}$, it follows that

$$(6.10) \quad G\Sigma = A'_H D_F^{-1} \Sigma = A'_H - \frac{1}{m} \underline{\alpha}_1 \underline{F}'$$

with $\underline{\alpha}_1$ the first column of A'_H .

Also,

$$(6.11) \quad G\Sigma G' = A'_H D_F^{-1} A_H$$

so that the estimate $\underline{f}_c$ of $\underline{F}_c$ in (6.4) reduces to

$$(6.12) \quad \underline{F}_c = \underline{F} - (A_H - \frac{1}{m} \underline{F} \underline{\alpha}'_1) (A'_H D_F^{-1} A_H)^* A'_H \ln \underline{F}.$$

The covariance matrix of the estimated frequencies under the hypothesis is given by

$$(6.13) \quad \Sigma_c = \Sigma - (A_H - \frac{1}{m} \underline{F}_c \underline{\alpha}'_1) (A'_H D_{F_c}^{-1} A_H)^* (A'_H - \frac{1}{m} \underline{\alpha}_1 \underline{F}'_c).$$

The estimators of the parameters in the log-linear model are

$$(6.14) \quad \hat{\underline{\lambda}}_c = (A'A)^{-1} A' \ln \underline{f}_c$$

with covariance matrix

$$(6.15) \quad (A'A)^{-1} A' D_{F_c}^{-1} \Sigma_c D_{F_c}^{-1} A (A'A)^{-1}.$$

Consider the following 2^2 contingency table as an example:

		Factor B	
		I	II
Factor A	I	$F_{11} = m$	F_{12}
	II	F_{21}	F_{22}

In this case the estimated frequencies under the hypothesis of independence are given by

$$(6.16) \quad \underline{F}_c = \underline{F} + e \begin{pmatrix} 0 \\ \frac{F_{12}}{m} + 1 \\ \frac{F_{21}}{m} + 1 \\ \frac{F_{22}}{m} - 1 \end{pmatrix}$$

where

$$e = \frac{(1, -1, -1, 1) \ln \underline{F}}{\frac{1}{F_{11}} + \frac{1}{F_{12}} + \frac{1}{F_{21}} + \frac{1}{F_{22}}}$$

The covariance matrix of $\underline{f}_c$ is

$$(6.17) \quad \Sigma_c = \Sigma - \frac{1}{\frac{1}{F_{11}} + \frac{1}{F_{12}} + \frac{1}{F_{21}} + \frac{1}{F_{22}}} \begin{pmatrix} 0 \\ -1 - \frac{F_{12}}{m} \\ -1 - \frac{F_{21}}{m} \\ 1 - \frac{F_{22}}{m} \end{pmatrix} \begin{pmatrix} 0, -1 - \frac{F_{12}}{m}, -1 - \frac{F_{21}}{m}, 1 - \frac{F_{22}}{m} \end{pmatrix}$$

6.3 A comparison of multinomial and negative multinomial sampling

In the case of multinomial sampling, where the total N is fixed, the expressions corresponding to (6.2), (6.12) and (6.13) are:

$$(6.18) \quad \Sigma = D_F - \frac{1}{N} \underline{F} \underline{F}'$$

$$(6.19) \quad \underline{F}_c = \underline{F} - A_H (A_H' D_F^{-1} A_H)^* A_H' \ln \underline{F}$$

$$(6.20) \quad \Sigma_c = D_{F_c} - \frac{1}{N} \underline{F}_c \underline{F}_c' - A_H (A_H' D_{F_c}^{-1} A_H)^* A_H'$$

The ratios of the estimated frequencies for multinomial and negative multinomial sampling are constant under the different hypotheses. For this reason the estimation of λ , with the exception of the mean μ , is the same for both the sampling models. As was already indicated in sections 3 and 4, the variances of the estimators differ however. Consider the example of a three-way contingency table given in Haberman (1978, p.183). Although the data was not collected according to a negative multinomial sampling scheme, the usual multinomial analysis is compared to a negative multinomial analysis (i) as if the first cell is the target cell and (ii) as if the 4th cell is the target cell. The relevant variables were gender (G), level of education (E) and attitude (A) with 2, 3 and 2 categories respectively. The comparison of the sampling models is given in Tables 6.1 and 6.2. As may be expected table 6.2 shows that for a relatively large predetermined count in the target cell the multinomial model and the negative multinomial model will in practical analyses lead to the same results.

TABLE 6.1
ESTIMATED FREQUENCIES FOR MODEL: $\lambda^{GEA} = 0$

<i>Variables GEA</i>	<i>Observed values</i>	<i>Estimated Multinomial</i>	<i>values Neg.Mult. (i)</i>	<i>Neg.Mult. (ii)</i>
111	28	35,7	28,0	35,4
112	187	179,3	140,7	177,6
121	173	170,4	133,6	168,8
122	283	285,6	224,1	283,0
131	86	80,9	63,5	80,2
132	38	43,1	33,8	42,7
211	44	36,3	28,5	36,0
212	179	186,7	146,4	185,0
221	110	112,6	88,4	111,6
222	196	193,4	151,7	191,6
231	72	77,1	60,4	76,3
232	47	41,9	32,9	41,6
	1443	1443,0	1132,0	1428,8

TABLE 6.2

INDEPENDENT λ -PARAMETER VALUES AND VARIANCES
ESTIMATED UNDER MODEL: $\lambda^{GOH} = 0$

λ - values		Variance	Neg. Mult (i)	Neg. Mult. (ii)
		Mult		
μ	4,5503	(Mult)	0,0004	
	4,3075	(Neg.Mult (i))	0,0194	0,0039
	4,5410	(Neg.Mult (ii))		
λ^A	0,0685		0,0011	0,0009
λ^B	-0,1540		0,0032	0,0025
	0,6450		0,0021	0,0016
λ^C	-0,2558		0,0014	0,0011
λ^{AB}	-0,0828		0,0023	0,0018
	0,1325		0,0017	0,0013
λ^{AC}	0,0059		0,0011	0,0009
λ^{BC}	-0,5571		0,0032	0,0025
	-0,0085		0,0020	0,0016

7. SIMULATIONS

In order to evaluate the normality assumptions empirically, a 4×5 negative multinomial two-way and a $2 \times 3 \times 4$ negative multinomial three-way table were simulated 5000 times for $m = 5$, $m = 10$ and $m = 50$. In the two-way case $\lambda^{AB} = 0$ and in the three-way case the base model was $\lambda^{ABC} = \lambda^{BC} = \lambda^{AC} = 0$. In Table 7.1 an illustration is given of the expected frequencies, averages of the observed and fitted frequencies and corresponding variances for one cell in the two way table and one cell in the three-way table. Graphical representations of the simulated distributions are given in Figure 7.1 and Figure 7.2. Table 7.2 gives the simulation results for one of the λ 's, viz. λ_{12}^{AB} , in the three-way table, while the corresponding simulated distributions are presented in Figure 7.3.

TABLE 7.1
RESULTS OF SIMULATIONS FOR CELL (2,1) AND CELL (2,3,3).

Cell m	Expected Frequencies	Average Frequency		Variance of Observed Frequencies		Variance of Fitted Frequencies	
		Observed	Fitted	True	Simulated	True	Simulated
(2,1)	5	6,55	6,53	6,59	15,1	15,1	2,5
	10	13,10	13,15	13,10	30,3	29,0	4,9
	50	65,50	65,50	65,50	151,3	150,5	24,5
(2,3,3)	5	9,48	9,53	9,56	27,5	27,5	9,6
	10	19,96	18,99	19,02	54,9	57,9	19,3
	50	94,82	95,25	94,97	274,7	272,5	96,3

The simulated results represented here are typical of all the simulated results. It seems that the assumptions of normality are reasonably justified, even for moderate values of m .

TABLE 7.2
RESULTS OF SIMULATIONS FOR λ_{12}^{AB} :
THREE-WAY TABLE

m	True value	Average simulated value		Variance observed		Variance fitted	
		Observed	fitted	True	Simulated	True	Simulated
5	0,23	0,230	0,218	0,0098	0,0126	0,0096	0,0102
10	0,23	0,230	0,233	0,0049	0,0058	0,0058	0,0049
50	0,23	0,230	0,229	0,0010	0,0010	0,0010	0,0010

REFERENCES

- BISHOP, Y.M.M., FIENBERG, S.E. and HOLLAND, P.W. (1975). *Discrete Multivariate Analysis*. Cambridge (Massachusetts): MIT Press.
- BONNETT, D.G. (1985). The negative multinomial logit model. *Comm. Statist.-Theor. Meth.*, **14**(7), 1713-1717.
- CRAMÉR, H. (1966). *Mathematical Methods of Statistics*. Princeton: Princeton Univ. Press.
- CROWTHER, N.A.S. and SHAW, T.M. (1989). *An estimation procedure in categorical data analysis*. South African Statist.J., **23**, 63-83.
- GLAS, A.P.(1986). *Loglineêre modellering in die geval van komplekse steekproefdata*. M.Sc.-thesis. University of Pretoria.
- GOKHALE, D.V. (1979). Analysis of ecological frequency data: certain case studies. In *Multivariate Methods in Ecological Work*, L. Orloci, C.R. Rao and W.M. Stiteler (Eds). Fairlands, Maryland: International Co-operative Publishing House.
- HABERMAN, S.J. (1978). *Analysis of Qualitative Data: Volume 1*. New York: Academic Press, Inc.
- JOHNSON, N.L. and KOTZ, S. (1969). *Discrete Distributions*. Boston: Houghton Mifflin Co.
- KENDALL, M.G. and STUART, A. (1959). *The Advanced Theory of Statistics*. London: Charles Griffin Co.
- MARSAGLIA, G. (1965). Ratios of normal variables and ratios of sums of uniform variables. *Journal of American Statistical Association.*, **10**, 192-204.
- NEYMAN, J. (1949). Contribution to the theory of the χ^2 test. *Proc. Berkeley Symp. Math. Statist. Prob.*, 239-272.

FIGURE 7.1
Simulated frequencies for cell (2,1)

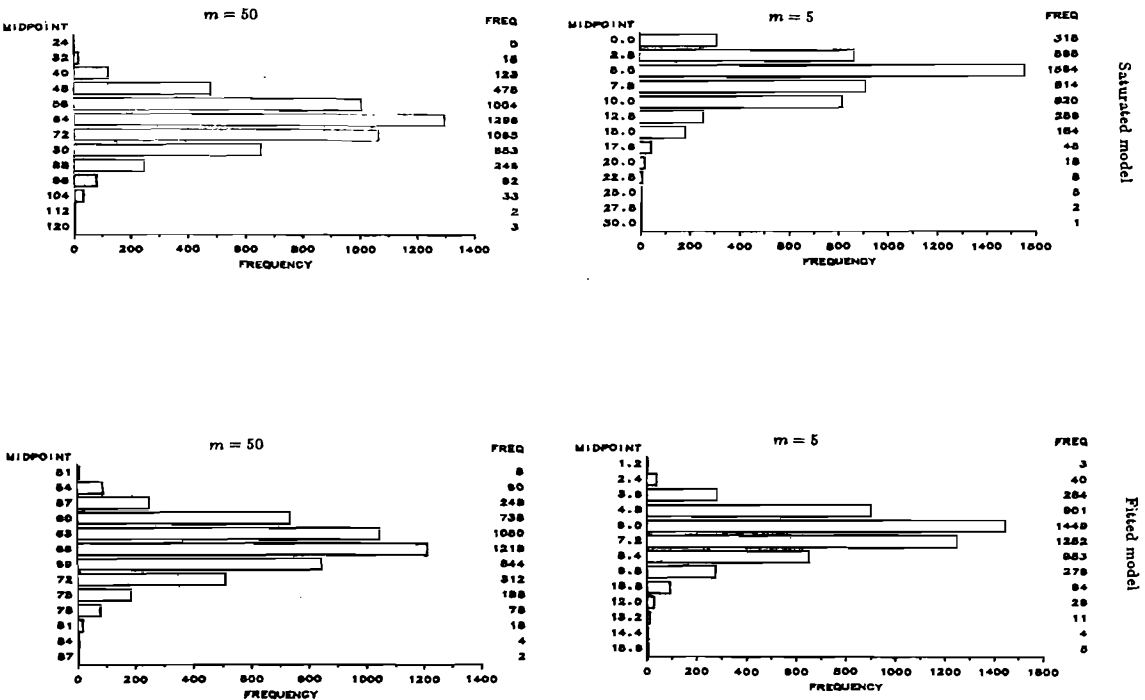


FIGURE 7.2
Simulated frequencies for cell (2,3,3)

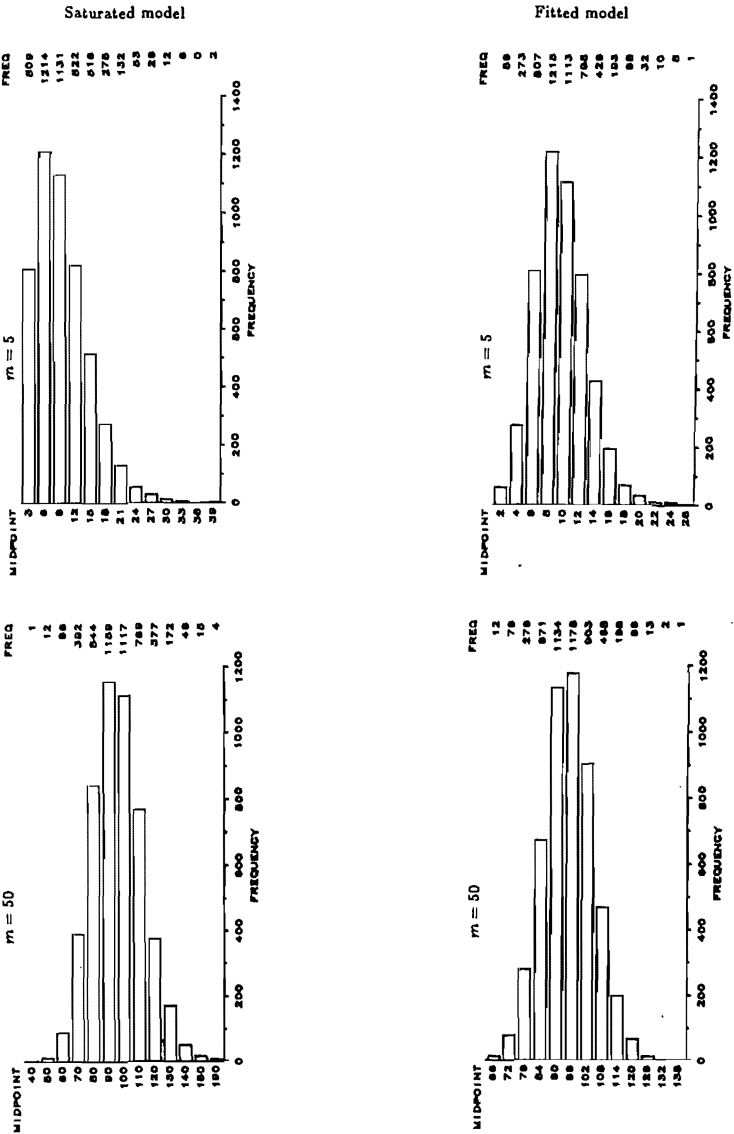
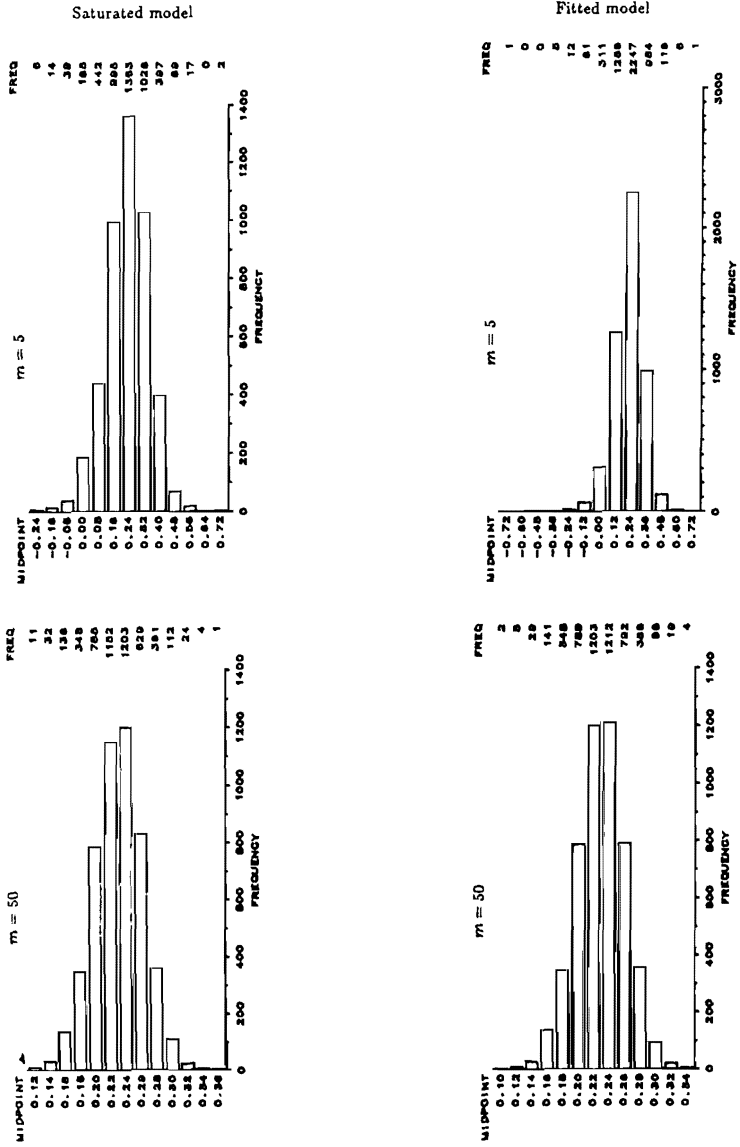


FIGURE 7.3
Simulated values for λ_{12}^{AB}



RUDOLPH, G.J. (1963). *Oor eksakte en asimptotiese benaderde toetse by geklassifiseerde gegewens*. Ph.D.-thesis. UNISA.

STEYN, H.S. (1955). On discrete multivariate probability functions of hypergeometric type. *Proc. koninkl. Nederl. Akad. van Wetenschappen*, Series A, 54, 3-10.

STEYN, H.S. (1959). On χ^2 -tests for contingency tables of negative binomial type. *Statistica Neerlandica*, 13, 433-444.

STEYN, H.S., CROWTHER, N.A.S., RAATH, E.L. and SHAW, T.M. (1987) 'n *Algemene model vir negatiewe multinomiale dataontleding*. Report WS-38. Pretoria: HSRC.

STOKER, D.J. and GLAS, A. (1984). *Die simulاسie van meerrigting-gebeurlikheidstabelle onder enige hiërargiese loglineêre model*. Occasional Publication No. 21. Pretoria: HSRC.

STUMPF, R.H. (1984). *Basic underlying multinomial models and related covariance structures in categorical data analysis*. Ph.D.-thesis. UNISA.

STUMPF, R.H. and STEYN, H.S. (1986). Exact distributions associated with an $I \times J \times K$ contingency table. *Comm. Statist.- Theor. Meth.*, 15(b), 1889-1904.

TAILLIE, C., ORD, J.K., MOSIMANN, J.E., PATIL, G.P. (1979). Chance mechanisms underlying multivariate distributions. In *Statistical Distributions in Ecological Work*, J.K. Ord, G.P. Patil and C. Taillie (Eds). Fairlands, Maryland: International Co-operative Publishing House.

YATCHEW, A. (1986). Multivariate distributions involving ratios of normal variables. *Comm. Statist. Theor. Meth.*, 15(6), 1905-1926.